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Summary of Savannah River Site FY23 Salt Waste Qualification Data

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EXECUTIVE SUMMARY

The Savannah River National Laboratory (SRNL) analyzed samples from Savannah River Site (SRS) Waste Tanks 41H and 21H to support qualification of Salt Waste Processing Facility (SWPF) Waste Batches 8 and 9 for processing (the FY23 Salt Batch Qualification samples). These Tanks (i.e. 41H and 21H) are blend tanks for feed to SWPF. None of the samples displayed any unusual or unexpected characteristics such as large amounts of solids, floating solids, or unusual color. Characterization of these samples confirmed similar chemical composition and characteristics to previous salt waste batches. The results for Batches 8 and 9 were provided by SRNL to Savannah River Mission Completion (SRMC), the Liquid Waste Operations subcontractor at SRS, as External Sample Results (Laboratory Information Management System (LIMS)) Reports. Additionally, a separate technical memo was issued by SRNL to report re-test data for Batch 8 for Cs-137 for filtered samples only which were run at the request of SWPF. For Batch 9, a set of samples was also analyzed in parallel by the SWPF-Analytical Laboratory (SWPF-AL). The SWPF-AL data is included herein for comparison with the SRNL data where applicable. The analytical results (both rapid, typically 4 weeks, and long term, typically 8 weeks) for Batches 8 and 9 are now summarized and discussed in this technical report.

SWPF Salt Batch 8 and 9 Qualification samples were similar in chemical composition to each other and to the previous 17 Salt Waste Batches (10 Interim Salt Waste Processing (ISWP) Macrobatch and 7 SWPF Salt Batch Qualification samples) analyzed to date with some results trending at or beyond the higher end of the ranges previously observed for some analytes. Salt Batch 8 and 9 results for sodium, aluminum, and hydroxide were on the high end or higher than results for previous salt batches analyzed to date. Salt Batches 8 and 9 contained 7.19 to 7.46 M Na^+ (the previously reported maximum value for all batches was 7.50 M Na^+ for Batch 6), 0.371 to 0.437 M AlO_2^- (the previously reported maximum value for all batches was 0.368 M AlO_2^- for Batch 6), 3.52 to 4.07 M OH^- (the previously reported maximum value was 3.78 M OH^- for Batch 6), 1.59 to 1.98 M NO_3^- (previous range 1.30 to 3.27 M NO_3^-), 0.87 to 1.16 M NO_2^- (previous range 0.22 to 1.28 M NO_2^-), 0.265 to 0.313 M CO_3^{2-} (previous range 0.11 to 0.39 M CO_3^{2-}), 0.033 to 0.042 M SO_4^{2-} (previous range 0.05 to 0.16 M SO_4^{2-}), and other minor chemical components. Total mercury concentrations ranged from 40.0 to 64.1 mg/L. These mercury values are in the middle of the range reported for all prior 17 Salt Batch Qualification samples (9.17 to 129.0 mg/L Hg).

Cs-137 activities for Batches 8 and 9 were as high or higher than the maximum reported (5.22E+08 pCi/mL, 1.16E+09 dpm/mL) for the previous 17 Salt Waste Batches analyzed to date. All Cs-137 activities, in both filtered and unfiltered samples of Salt Batches 8 and 9, ranged from 4.78E+08 pCi/mL (1.1E+09 dpm/mL, 1.8 Ci/gal) to 6.22E+08 pCi/mL (1.4E+09 dpm/mL; 2.4 Ci/gal). The Cs-137 activity for Salt Batch 8 filtered sample was higher, 6.12E+08 pCi/mL, than the previous maximum reported 5.2E+08 pCi/mL (1.2E+09 dpm/mL; 2.0 Ci/gal) which was for filtered Salt Batch 6. Revisions continue to be made to the SWPF Waste Acceptance Criteria (WAC) requirement related to Cs-137 content. During the time frame of these qualification samples, the SWPF WAC for Cs-137 content was raised to 5.1 Ci/gal but has subsequently been lowered to 3.0 Ci/gal (uncertainty 2 sigma applies) based on SWPF Safety Basis calculations related to hydrogen generation where Cs-137 value is an input.

Sample filtration had little impact on the total gamma and beta activities for the samples indicating that these activities were primarily associated with soluble species. Comparing filtered versus unfiltered samples, both Batch 8 and Batch 9 samples had higher activity in the unfiltered sample as compared to the filtered sample for Cs-137 but not significantly higher (SRNL estimated 1σ as 5 %). This trend was the same for most of the other radionuclide analyses as well. The total beta activity was similar to the Cs-137 values measured. For filtered Batch 8 the total beta activity was 6.17E+08 pCi/mL (1.4E+09 dpm/mL; 2.3 Ci/gal) and was only slightly higher than the filtered Cs-137 measured value 6.12E+08 pCi/mL (re-test value for four replicates). For filtered Batch 9 the total beta activity was 5.8E+08 pCi/mL (1.3E+09 dpm/mL; 2.2 Ci/gal) which was only 20% higher than the Cs-137 (primary beta emitter) activities

for this sample. The total gamma activities for the filtered samples for Batch 8, 5.66E+08 pCi/mL (1.3E+09 dpm/mL; 2.14 Ci/gal) are higher than Batch 9, 4.62E+08 pCi/mL (1.0E+09 dpm/mL, 1.75 Ci/gal), as well as Batch 6, 4.94E+08 pCi/mL (1.1E+09 dpm/mL; 1.87 Ci/gal).

As compared to the previous 17 Batches, the trends following cesium removal are similar for Batches 8 and 9. Total beta activity for the samples decreased dramatically (two orders of magnitude) to 1.32E+06 pCi/mL (2.9E+06 dpm/mL, 5.0E-03 Ci/gal) for filtered Batch 8 and 9.59E+05 pCi/mL (2.1E+06 dpm/mL, 3.6E-03 Ci/gal) for filtered Batch 9 as compared to Cs-137 values. Activities for radionuclides other than Cs-137 and Ba-137m in the samples were primarily associated with Sr-90/Y-90, Tc-99, Pu-238 and Pu-241. Although I-129 activities are relatively low in comparison to the other radionuclides reported herein, the measured value for Batch 8 was higher (167 pCi/mL) than Batch 9 (93.8 pCi/mL) and was also higher than the highest value reported for Salt Batches 1-7 (107 pCi/mL (Batch 6)). Although I-129 values did not exceed the SWPF WAC and do not impact the qualification of these Salt Batches or their processing through SWPF, additional measures are in place to reevaluate I-129 prior to transferring to Tank 49 as a part of an active deviation for Salt Batches 8-22 to justify exceeding the Saltstone Production Facility (SPF) WAC limit if it is not met at that time.

The plutonium isotopes Pu-238, Pu-239 and Pu-240 are the major contributors to the alpha activity in SRS Salt Batch Feed samples, and Pu-238 accounts for the majority of the alpha activity in Salt Batch Samples 8 and 9. For both Batch 8 and Batch 9 analyzed by SRNL, the Pu-238 activities of the unfiltered samples were higher than the filtered sample by a significant amount (greater than 2σ estimated at 10% for SRNL analyses) indicating that not all of the Pu-238 is soluble for these batches. This is a departure in behavior from previous Salt Batches. Other Pu isotopes although much lower in activity than Pu-238 also exhibited similar behavior (Pu-239 (Batches 8 and 9) and Pu-240 (Batch 8)). Pu-238 activity considering both filtered and unfiltered samples ranged in value from 6.41E+03 pCi/mL (1.4E+04 dpm/mL) for Batch 9 (filtered) to 4.52E+04 pCi/mL (1.0E+05 dpm/mL) for Batch 8 (unfiltered). The maximum Pu-238 activity considering only filtered samples in Batches 8 and 9 was 2.99E+04 pCi/mL (6.6E+04 dpm/mL) (Batch 8). The maximum Pu-238 values reported previously were 8.39E+04 pCi/mL (1.9E+05 dpm/mL) for filtered (Batch 5), and 9.14E+04 pCi/mL (2.0E+05 dpm/mL) for unfiltered (Batch 5). These activities are at or near the values for the total alpha with cesium removed method. The total alpha measurement following cesium removal is a gross semi-quantitative analysis (suffering from the high beta to alpha activity ratio in Salt Batch matrices) and has poor precision compared to the sum of the quantitative radiochemical analyses results for the various actinides.

Comparisons of the chemical and radionuclide compositions of Salt Waste Processing Facility Batches 8 and 9 to all of the previous 17 batches have for the most part exhibited similar compositions. Data, where available, reported by SWPF-AL for Batch 9 was similar to data reported by SRNL with SWPF-AL data trending slightly lower for most analyses reported. The major anion concentrations and the radionuclide concentrations for the two batches were similar with a number of analytes generally trending higher as compared to previously reported Salt Batch data for the previous 17 Salt Batch samples. Although specific analytes, including sodium, aluminum, and hydroxide, exceeded targets included in the SWPF WAC and I-129 exceeded the SPF WAC limit, no significant impact was deduced from the results based on evaluations carried out to justify any WAC compliance.

Note: The SWPF WAC has undergone multiple recent revisions before, during, and after FY23 including changes such as glycolic flowsheet addition and other adjustments including sodium and cesium concentrations. This report focuses on the characterization of the Salt Batch Qualification sub-samples. Prior to processing for cesium removal, the Qualification Batches were transferred to process feed tanks which contained the heel from the previous Salt Batch. In addition, in some cases NaOH reagent was added to adjust the waste free hydroxide. These activities resulted in minor changes to the final waste compositions processed. The final compositions were estimated based on these blending and adjustment activities and the calculated compositions are provided in various Liquid Waste Operations subcontractor reports which are not summarized in this document.

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LIST OF ABBREVIATIONS

ARP	Actinide Removal Process
CSTF	Concentration, Storage, and Treatment Facilities
DSS	Decontaminated Salt Solution
DWPF	Defense Waste Processing Facility
IC	Ion Chromatography
ICPES	Inductively Coupled Plasma Emission Spectroscopy
IDP	Inhalation Dose Potential
ISWP	Interim Salt Waste Processing
LIMS	Laboratory Information Management System
MCU	Modular Caustic Side Solvent Extraction Unit
MST/SS	Monosodium Titanate/Sludge Solids
SRMC	Savannah River Mission Completion
SRNL	Savannah River National Laboratory
SRS	Savannah River Site
SWPF	Salt Waste Processing Facility
SWPF-AL	Salt Waste Processing Facility-Analytical Laboratory
SPF	Saltstone Production Facility
TIC/TOC	Total Inorganic Carbon/Total Organic Carbon
VOA/SVOA	Volatile Organic Analysis/Semivolatile Organic Analysis
TTR	Technical Task Request
TTQAP	Task Technical and Quality Assurance Plan
WAC	Waste Acceptance Criteria

1.0 Introduction

This report provides a summary of the FY23 Salt Batch Qualification efforts and a comparison of those results with the previous Salt Batch Qualification results. SRNL analyzed two Salt Batch Waste samples from SRS, Concentration, Storage, and Transfer Facility Tank Farms Waste Tanks 41H and 21H (Salt Batches 8 and 9, respectively). SWPF-AL also performed short term analyses on Tank 21H (Salt Batch 9) samples. This report details all results from SRNL analyses performed on these qualification samples alongside the results from SWPF-AL analyses where available. In past years, numerous Salt Waste Batch sub-samples have been retrieved, and this data is summarized for 17 prior Salt Waste Batches (7 Salt Batch and 10 Macrobatch Qualification samples) in the previous FY21 and FY22 Summary documents.[i,ii] Salt Batches were processed at SWPF, whereas the Macrobatches were processed through the Interim Salt Waste Processing (ISWP) system which included the Actinide Removal Process (ARP) and the Modular Caustic-Side Solvent Extraction Unit (MCU).

As described in the waste acceptance criteria (WAC) document,[iii] the SWPF Feed acceptance criteria ensure that the monosodium titanate/sludge solids (MST/SS) and strip effluent streams from SWPF will not exceed applicable Defense Waste Processing Facility (DWPF) safety and processing constraints (e.g., canister heat generation, nuclear criticality safety, corrosive species, and organic concentration). The SWPF Feed acceptance criteria also ensures that the decontaminated salt solution (DSS) stream from SWPF will not exceed applicable Tank 50H and SPF safety and processing constraints (e.g., inhalation dose potential (IDP)), nuclear criticality safety (fissile material and fissile poisoning), radiolytic hydrogen generation, and organic concentration.

A Task Technical and Quality Assurance Plan (TTQAP)[iv] requests SRNL to provide annual technical reports of the compiled Salt Batch Qualification data to Savannah River Mission Completion (SRMC), the current Liquid Waste Operations Sub-contractor.

2.0 Experimental

2.1 Sample Receipt and Preparation

Samples were retrieved on 11/13/2022 from SRS Tank 41H for characterization of SWPF Salt Batch 8. The samples were transferred from H-Tank farm and received by SRNL on 11/14/2022 for characterization as described in **Table 2-1**. Samples of Salt Batch 9 were analyzed by both SRNL and SWPF-AL. Samples were retrieved from SRS Tank 21H for characterization of SWPF Salt Batch 9 as described in **Table 2-1**. The details of batch compositions, mixing and sampling for these Salt Batches have been documented in X-ESR-H-01164 Rev 2.[v]

The Salt Batch 8 qualification sample was a composite of four 200-mL tank samples listed in **Table 2-1** which were combined into one 800-mL sample. SRNL performed all analyses for the Salt Batch 8 Qualification sample.

The Salt Batch 9 Qualification sample consisted of two sets of composite samples **Table 2-1**. The first set of four 200-mL samples was pulled on 5/15/2023 and delivered to SRNL on 5/16/2023. These samples were combined into a single 800-mL sample. The second set of four 200-mL samples was pulled on 5/16/2023 and delivered to the SWPF-AL on 5/17/2024. SRNL performed both short- and long-term analyses on the first sample of Salt Batch 9, and SWPF-AL performed short term analyses on the second sample. SWPF-AL results are included for comparison with SRNL short term results. The specific details of the collection of the SWPF-AL data are not included herein. Batch 9 is the first Salt Batch that SWPF-AL has analyzed samples associated with Salt Batch Qualification.

Table 2-1. Salt Batch Sample Description and Report References

SWPF Batch	Qualification Tank	Sample	References
8	41H	HTF-41-22-109/110/111/112 (composite 800 mL)	LW-AD-PROJ-221107-1[vi] SRNL-L3120-2023-00005[vii]
9	21H	HTF-21-23-36/37/38/39 (composite 800 mL)	LW-AD-PROJ-230331-4[viii]
		HTF-21-23-40/41/42/43 (composite 800 mL)	AFR-1007530, Rev 2[ix]

For the analyses conducted in SRNL, the details of the analyses are as follows. Density was measured using a 2-mL flask at ambient temperature in the SRNL shielded cells, and the temperature was recorded. The densities of filtered (0.45-µm filter) and unfiltered sub-samples of each batch were determined in duplicate, and the average result was reported. For the total mercury and methyl mercury measurements, , approximately 1.5-mL sub-samples were diluted into 40 mL of ultrapure water acidified with 0.2 mL of ultrapure HCl prior to adding the 1.5 mL of sub-sample. Dilution was carried out in Teflon-capped, glass vials with very little headspace and the diluted samples were removed from the shielded cells and stored below 4 °C (in a refrigerator), following the recommended method for preparing samples of this type.[x] To measure the insoluble solids content (weight %) , approximately 300 mL of the well-mixed salt solution was weighed on an electronic balance and transferred through a 0.2-µm porosity nylon filter to collect the insoluble solids. The solids and filter were washed with several ~50-mL portions of deionized water to remove the soluble salts. The solids were then dried to constant mass. For other unfiltered samples, a well-mixed solution was removed from the shielded cells without dilution or digested in acid (specifically aqua regia) in the shielded cells prior to analysis. For samples requiring filtration and containing visible settling solids, filtration was conducted using a 0.45-µm nylon syringe filter and the filtrate was submitted for analysis.

2.2 Quality Assurance

Requirements for performing reviews of technical reports and the extent of review are established in Manual E7 2.60. SRNL documents the extent and type of review using the SRNL Technical Report Design Checklist contained in WSRC-IM-2002-00011, Rev. 2. The work performed, all analyses, and the review process for this report complies with those requirements. Results from this report and those reports for which results are compiled and summarized are not governed by DOE/RW-0333P (enhanced Quality Assurance requirements) as per the Technical Task Request (TTR).[xi] Design verification of this report was performed by Document Review.

3.0 Characterization Results for the FY23 SWPF Salt Batches (8 and 9) including analyses by SRNL and SWPF-AL

SRNL analyzed samples for major ions by ion chromatography (IC) (formate, nitrate, nitrite, sulfate, phosphate, oxalate, chloride and fluoride), inductively coupled plasma-emission spectrometry (ICP-ES) (Al, Na, K, and Si), and total inorganic carbon/total organic carbon analysis (TIC/TOC) (Carbonate and TOC). Total mercury and methyl mercury analyses were also performed. A number of additional semi-volatile organic analysis (SVOA) (Tributyl phosphate and all other SVOA compounds) as well as volatile analysis (VOA) compounds were analyzed including butanol and all other VOA compounds. None of these analyses for VOA/SVOA were measured above the detection limits and therefore are not included in **Table 3-1**.

Table 3-1. SWPF Salt Batch 8 and 9 Density (g/mL) and Composition (mg/L (M)).

SWPF Batch	**8[a]**	**9[a]**	**9[a]**
	SRNL	**SRNL**	**SWPFAL**
Density	**g/mL**	**g/mL**	**g/mL**
	1.31	1.27	---
	1.32[b]	1.30[b]	---
Insoluble solids	**mg/L**	**mg/L**	**mg/L**
	184	4.48	**<200**
Component	**mg/L (M)**	**mg/L (M)**	**mg/L (M)**
Na^+	1.70E+05 (7.41)	1.72E+05 (7.46)	1.65E+05 (7.19)
Al	1.18E+04 (0.437)	1.07E+04 (0.397)	1.00E+04 (0.371)
K^+	1.60E+03 (0.0409)	1.26E+03 (0.0322)	1.21E+03 (0.0310)
Si	<2.74E+01 (<0.000976)	<3.99E+01 (<0.00142)	8.00E+01 (0.00285)
OH^-	6.93E+04 (4.07)	5.98E+04 (3.52)	6.00E+04 (3.53)
NO_3^-	9.83E+04 (1.59)	1.16E+05 (1.87)	1.23E+05 (1.98)
CO_3^{2-}	1.59E+04 (0.265)	1.88E+04 (0.313)	1.70E+04 (0.283)
NO_2^-	5.33E+04 (1.16)	4.26E+04 (0.926)	3.98E+04 (0.865)
SO_4^{2-}	4.00E+03 (0.0416)	3.23E+03 (0.0336)	3.12E+03 (0.0325)
PO_4^{3-}	4.47E+02 (0.00471)	3.53E+02 (0.00372)	4.47E+02 (0.00471)
$C_2O_4^{2-}$	1.02E+02 (0.00116)	<1.00E+02 (<0.00114)	9.50E+01 (0.00108)
HCO_2^-	<1.00E+02 (<0.00222)	<1.00E+02 (<0.00222)	<1.00E+01 (<0.00022)
F^-	<1.00E+02 (<0.00526)	<1.00E+02 (<0.00526)	3.15E+01 (0.00166)
Cl^-	3.05E+02 (0.00860)	2.54E+02 (0.00716)	2.39E+02 (0.00674)
Total Hg	6.41E+01 (0.000320)[b]	5.04E+01 (0.000251)[b]	4.00E+01 (0.000199)[b]
Methyl Hg	8.88E+00[b]	9.35E+00[b]	---
Total Organic Carbon	1.06E+02	7.17E+01	1.20E+02

Note: molar values (M) calculated from reported values (mg/L) except for sodium in which case molar concentration (M) was the reported value.
[a] filtered unless otherwise indicated and
[b] unfiltered
--- not requested

Radioisotopes analyzed with results determined to be above the instrument detection limit are listed in **Table 3-2**. A number of additional radionuclide analyses were carried out as specified in the TTQAP.[iv] Those radionuclides below the limit of detection for Salt Batch 8 and 9 were not included in this summary report. Analytes measured but not included in **Table 3-2** are Am-241, Cm-245, Ce-144, Co-60, Eu-154, Eu-155, and Sb-125. Detection limits for all analyses were below the Waste Acceptance Criteria (WAC) limits.

SRNL utilized automated LIMS data reporting for the initial reporting of the data.[vi,viii] A re-test was requested of SRNL for Salt Batch 8 to remeasure Cs-137 in filtered samples starting with new aliquots of original sample. This re-test was reported in a separate document.[vii] The re-test data reported herein is an average of all four values measured (two original and two re-test values). The average Cs-137 value reported in the re-test[vii] was used in the evaluation of Salt Batch 8 instead of the value reported in **LW-AD-PROJ-221107-1**.[vi] For Batch 9, both SRNL and SWPF-AL data are show in **Table 3-2**. SRNL data was used in the evaluation included in the Acceptance Report for Salt Batch 9.

Table 3-2. SWPF Salt Batch 8 and 9 Radionuclide Activities

SWPF Batch	8(filtered)	8(unfiltered)	9(filtered)	9(unfiltered)	9(filtered)	9(unfiltered)
Analyte	**SRNL**	**SRNL**	**SRNL**	**SRNL**	**SWPF-AL**	**SWPF-AL**
Units	**pCi/mL**	**pCi/mL**	**pCi/mL**	**pCi/mL**	**pCi/mL**	**pCi/mL**
^{137}Cs	5.99E+08	6.22E+08	4.88E+08	5.20E+08	4.78E+08	---
^{137}Cs Re-test[b]	6.12E+08	---	---	---	---	---
^{137m}Ba	5.66E+08	5.88E+08	4.62E+08	4.92E+08	---	---
Total gamma	5.66E+08	5.88E+08	4.62E+08	4.92E+08	---	4.08E+08
Beta Count (LSC[a])	6.17E+08	6.16E+08	5.81E+08	5.97E+08	---	5.72E+08
Total beta, Cs removed	1.32E+06	2.18E+06	9.59E+05	1.27E+06	1.04E+06	1.16E+06
Total alpha, Cs removed	4.16E+04	8.61E+04	<1.95E+04	<1.35E+05	6.10E+03	5.70E+03
^{238}Pu	2.99E+04	4.52E+04	6.41E+03	1.15E+04	---	7.30E+03
^{239}Pu	5.04E+02	8.20E+02	<4.50E+02	6.05E+02	4.00E+02	3.90E+02
^{240}Pu	2.76E+02	4.03E+02*	< 6.66E+02	2.78E+02	1.18E+02	1.22E+02
^{241}Pu	1.22E+04	---	7.54E+03	---	---	---
^{242}Pu	<1.05E+00	<6.69E+00	<1.16E+01	<5.74E-01	---	1.77E+00
^{244}Pu	<4.91E-03	<3.10E-02	<5.36E-02	<2.67E-03	---	2.3E-01
^{233}U	<1.93E+01	<1.85E+01	<2.41E+01	<1.20E+02	8.9E+00	6.4E+00
^{234}U	5.92E+01	6.12E+01	3.43E+01	<7.73E+01	3.1E+01	2.8E+01
^{235}U	1.39E-01	1.47E-01	2.49E-01	2.47E-01	<2.0E-01	2.17E-01
^{236}U	9.79E-01	9.80E-01	9.72E-01	8.82E-01	8.4E-01	7.9E-01
^{238}U	1.13E+00	1.23E+00	2.07E+00	2.00E+00	1.92E+00	1.74E+00
^{90}Sr	3.69E+05	5.40E+05	2.84E+05	7.03E+05	3.42E+05	3.82E+05
^{90}Y	3.69E+05	5.40E+05	2.84E+05	7.03E+05	---	---
^{3}H	3.04E+03	---	5.65E+03	---	---	---
^{14}C	4.95E+03	---	<9.32E+02	---	---	---
^{99}Tc	1.63E+05	---	1.23E+05	---	---	---
^{126}Sn	2.16E+03	---	1.74E+03	---	---	---
^{129}I	1.67E+02	---	9.38E+01	---	---	---
^{237}Np	5.99E+00	---	6.21E+00	---	---	---
^{244}Cm	3.10E+01	---	<9.90E-01	---	---	---

[a] liquid scintillation counting (LSC)

[b] re-test data reported is an average including both original measurements and additional measurements (4 replicates total)

--- not requested

* replicates include values that were below detection

4.0 Comparison with Previous Salt Batch Characterization Results

SWPF Salt Batch qualification samples 8 and 9 were similar in chemical composition to each other and to the previous 17 salt batches analyzed to date with results trending similar to or on the higher end of the ranges previously observed for most analytes. Batches analyzed prior to Salt Batches 8 and 9 include ISWP Salt Macrobatches 1-10 and SWPF Salt Batches 1-7.[i,ii] SWPF-AL data reported for Salt Batch 9 is, for the most part, in agreement with data reported by SRNL.

Results that were similar to previous Salt Batches included densities and insoluble solids. For Salt Batch 8 and 9 densities remained very close to 1.3 g/mL and the values measured for insoluble solids remained less than 200 mg/L.

Salt Batch 8 and 9 results for sodium, aluminum, and hydroxide were as high (within the instrument uncertainty; typically 10% (1σ as reported by SRNL) for these specific analyses) or higher than results for salt batches analyzed to date. Salt Batches 8 and 9 contained 7.19 to 7.46 M Na^+ (the previously reported maximum value for all batches was 7.50 M Na^+ for Batch 6), 0.371 to 0.437 M AlO_2^- (the previously reported maximum value for all batches was 0.368 M AlO_2^- for Batch 6), 3.52 to 4.07 M OH^- (the previously reported maximum value was 3.78 M OH^- for Batch 6),[ii] 1.59 to 1.98 M NO_3^- (previous range 1.30 to 3.27 M NO_3^-), 0.87 to 1.16 M NO_2^- (previous range 0.22 to 1.28 M NO_2^-), 0.265 to 0.313 M CO_3^{2-} (previous range 0.11 to 0.39 M CO_3^{2-}), 0.033 to 0.042 M SO_4^{2-} (previous range 0.05 to 0.16 M SO_4^{2-}) and other minor chemical components. See **Figure 4-1** for sodium and anion molar concentrations including SRNL data for Batches 5 through 9 and SWPFAL data for Batch 9. Total mercury concentrations ranged from 40.0 to 64.1 mg/L. These mercury values are in the middle of the range reported for all prior 17 Salt Batches (9.17 to 129.0 mg/L Hg).[i,ii] There were a number of analyses requested as a part of the TTQAP[iv] that were not measured at values above the instrument detection limit and are not included in this report. These analyses include SVOA and VOA.

Sample filtration had little impact on the total gamma and beta activities for the samples indicating that these activities were primarily associated with soluble species. All Cs-137 activities, in both filtered and unfiltered samples of Salt Batches 8 and 9, ranged from 4.78E+08 pCi/mL (1.1E+09 dpm/mL, 1.8 Ci/gal) to 6.22E+08 pCi/mL (1.4E+09 dpm/mL; 2.4 Ci/gal). The Cs-137 activity for Salt Batch 8 for the filtered sample was higher, 6.12E+08 pCi/mL, than the previous maximum reported 5.2E+08 pCi/mL (1.16E+09 dpm/mL; 2.0 Ci/gal) which was for filtered Salt Batch 6. Comparing filtered versus unfiltered samples, both Batch 8 and Batch 9 samples had higher activity in the unfiltered sample as compared to the filtered sample for Cs-137 but not significantly higher as compared to the uncertainty of the measurement (typically 1σ estimated at 5% for Cs-137 by SRNL). This observation was the same for the other radionuclide analyses as well. The total beta activity was similar to the Cs-137 values measured. For filtered Batch 8 the total beta activity was 6.17E+08 pCi/mL (1.4E+09 dpm/mL; 2.3 Ci/gal) which was higher but not significantly higher than the filtered Cs-137 measured value 6.12E+08 pCi/mL (re-test value for four replicates). For filtered Batch 9 the total beta activity was 5.8E+08 pCi/mL (1.3E+09 dpm/mL; 2.2 Ci/gal) which was 20% higher than the Cs-137 (primary beta emitter) activities for this sample. Ba-137m activities were calculated from the cesium data by multiplying the Cs-137 activities by 0.946 based on the fact that 94.6% of the Cs-137 decays to Ba-137m by beta decay and a secular equilibrium exists between these isotopes. The total gamma reported is the calculated Ba-137m activity for each sample, since Ba-137m is the dominant gamma emitter in the samples. The total gamma activities for the filtered samples for Batch 8, 5.66E+08 pCi/mL (1.3E+09 dpm/mL; 2.14 Ci/gal) are higher than Batch 9, 4.62E+08 pCi/mL (1.0E+09 dpm/mL, 1.75 Ci/gal), as well as Batch 6, 4.94E+08 pCi/mL, (1.1E+09 dpm/mL; 1.87 Ci/gal). Revisions continue to be made to the SWPF Waste Acceptance Criteria (WAC) requirement related to Cs-137 content. Around the time of these qualification samples the SWPF WAC for Cs-137 content had been raised to 5.1 Ci/gal but has subsequently been lowered to 3.0 Ci/gal (uncertainty 2 sigma applies) based on SWPF Safety Basis calculations related to hydrogen generation where Cs-137 value is an input.[xii]

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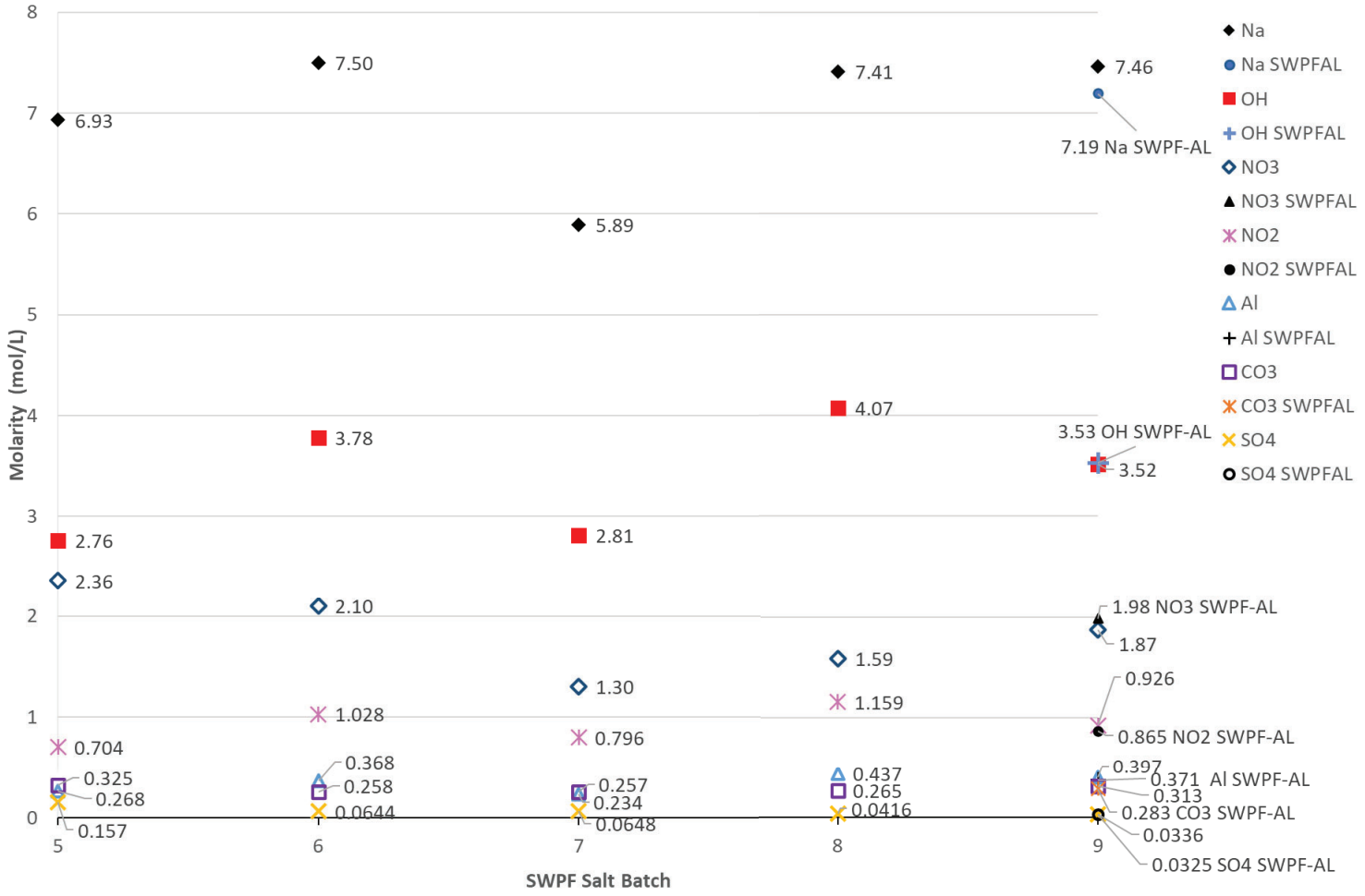


Figure 4-1. Sodium and Anion Molar Concentrations for Qualification Samples (All SRNL Salt Batch 5 through 9 Data and SWPF-AL Salt Batch 9 Data Included for Comparison)

As compared to the previous 17 Batches, the trends following cesium removal are similar for Batch 8 and 9. Total beta activity for the samples decreased dramatically (two orders of magnitude) to 1.32E+06 pCi/mL (2.9E+06 dpm/mL, 5.0E-03 Ci/gal) for filtered Batch 8 and 9.59E+05 pCi/mL (2.1E+06 dpm/mL, 3.6E-03 Ci/gal) for filtered Batch 9 as compared to Cs-137 values. Activities for radionuclides other than Cs-137 and Ba-137m in the samples were primarily associated with Sr-90/Y-90, Tc-99, Pu-238, and Pu-241. Although I-129 activities are relatively low in comparison to the other radionuclides reported herein, the measured value for Batch 8 was higher (167 pCi/mL) than Batch 9 (93.8 pCi/mL) and was also higher than the highest value reported for Salt Batches 1-7 (107 pCi/mL (Batch 6)). Although I-129 values did not exceed the SWPF WAC and do not impact the qualification of these Salt Batches or their processing through SWPF, additional measures are in place to reevaluate I-129 prior to transferring to Tank 49 as a part of an active deviation for Salt Batches 8-22 to justify exceeding the SPF WAC limit if it is not met at that time.

Pu is the primary alpha emitter in tank waste. The plutonium isotopes Pu-238, Pu-239, and Pu-240 are the major contributors to the alpha activity in SRS Salt Batch Feed samples, and Pu-238 accounts for the majority of the alpha activity in Salt Batch Samples 8 and 9. For both Batch 8 and Batch 9, the Pu-238 activity of the unfiltered sample was higher than the filtered sample by a significant amount indicating that not all of the Pu-238 is soluble for these Salt Batches (8 and 9). This is a departure in behavior from previous Salt Batches. Other Pu isotopes, although much lower in activity than Pu-238, also exhibited this behavior (Pu-239 (Batches 8 and 9, SRNL) and Pu-240 (Batch 8)). Pu-238 activity, considering both filtered and unfiltered samples, ranged in value from 6.41E+03 pCi/mL (1.4E+04 dpm/mL) for Batch 9 (filtered) to 4.52E+04 pCi/mL (1.0E+05 dpm/mL) for Batch 8 (unfiltered). The maximum Pu-238 activity considering only filtered samples in Batches 8 and 9 was 2.99E+04 pCi/mL (6.6E+04 dpm/mL) (Batch 8). The maximum Pu-238 values reported previously[i] were 8.39E+04 pCi/mL (1.9E+05 dpm/mL) for filtered (Batch 5) and 9.14E+04 pCi/mL (2.0E+05 dpm/mL) for unfiltered (Batch 5). These activities are at or near the values for the total alpha with cesium removed method. The total alpha measurement following cesium removal is a gross semi-quantitative analysis (suffering from the high beta to alpha activity ratio in Salt Batch matrices) and has poor precision compared to the sum of the quantitative radiochemical analyses results for the various actinides.

5.0 Conclusions

None of the samples displayed any unusual or unexpected visual characteristics. In general, the FY23 Salt Batch Samples analyzed in Batches 8 and 9 were similar to each other in chemical and radiological composition and followed similar trends for filtered, unfiltered, and samples with cesium removal prior to analysis. A significant difference in unfiltered and filtered values was observed, however, for Pu-238 and some other Pu isotopes. This observation indicates that some of the Pu species in Batches 8 and 9 may be insoluble. For Batch 9 analyses, the data reported by SWPF-AL was in reasonable agreement with the values reported by SRNL. SWPF-AL data values trended slightly lower than SRNL measured values for most analyses. Sample filtration had little impact on the total gamma and beta activities for the samples indicating that these activities were primarily associated with soluble species. Values for a number of analytes measured in Batches 8 and 9 continued to trend toward values as high or higher than previously reported for the 17 previous Salt Batches (10 Macrobatch samples and 7 Salt Batch Samples). These analyses trending higher included sodium, aluminum, and hydroxide as well as Cs-137 (and related analyses for Ba-137m and total gamma). Although the concentrations and activities were higher in value and exceeded targets included in the SWPF WAC for sodium, aluminum, and hydroxide and the SPF WAC limit for I-129, no significant impact was deduced from the results based on evaluations carried out to justify any WAC compliance.

6.0 Recommendations, Path Forward or Future Work

In the next summary document, the average values and ranges for key measured analytes for all Salt Batches reported to date through Salt Batch 9 (10 Macrobatch plus 9 Salt Batch samples) should be updated in order to be compared with the FY24 Salt Batch sample data. A methodology has not yet been fully developed for addressing final calculated compositions of the wastes processed through the ISWP facility and the SWPF which takes into account the impacts of the process feed tank heel compositions and added NaOH reagent. Once developed, this methodology should be considered in future revisions of this report. Also, a comparison of the final blended Salt Batch feed composition and the corresponding Tank 50H decontaminated salt solution may be considered to complement future summaries.

7.0 References

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